

CHAPTER 2.6 - HOLONS: THE UNIVERSAL BUILDING BLOCKS OF LIFE

1. The Parable of the Two Watchmakers

Allow me to introduce Tempus and Hora. Two watchmakers making identical watches of a thousand pieces. Tempus builds his watches in one go, and he does so very efficiently. But when he is interrupted – for example by a phone call from a customer – his entire assembly falls apart on the bench and he has to start all over again.

Hora has a different strategy. He first assembles sub-assemblies of ten pieces. Then he assembles those into larger sub-assemblies of a hundred pieces. And finally he assembles ten of those into a watch. Making these sub-assemblies takes a little bit longer per watch. But when the phone rings he only loses whatever module he happens to be working on. Everything already assembled holds its shape.

As it turns out, you can prove mathematically that in almost all complex systems, following the method of Hora – using sub-assemblies – makes you both more efficient and more resilient. It can be applied to almost anything, and we are going to apply it to the electricity grid.

The current grid, the grid we inherited, is like Tempus. When a substation on the Iberian Peninsula lost sync in April 2025 and all the lights went out from Lisbon to Marseille, it was like Tempus dropping a watch. It was an example of how fragile monolithic systems can be. Such systems become much more resilient by nesting them – by giving them sub-assemblies inside sub-assemblies.

In this chapter we dive into the theory. And we give it a name: *holons*.

2. Herbert Simon and Near-Decomposability

The man who came up with the parable of the two watchmakers was [Herbert Simon](#).

Simon is the kind of all-round genius who would make Leonardo Da Vinci jealous. He won the [Nobel Prize in Economics](#) in 1978 and, three years earlier, the [Turing Award](#) for artificial intelligence – the only person in history to have won both. He helped found Carnegie Mellon's business school and its computer science department. He was an amateur pianist, painter and mountain climber, and his professional career ran through economics, political science, cognitive psychology, and he became the father of artificial intelligence. In the year before he won the Nobel he was, for reasons of personal interest, teaching an undergraduate course on the French Revolution.

By the end of his life he could read professional papers in roughly twenty languages and literature for pleasure in half a dozen. He taught himself Dutch specifically in order to read Adriaan de Groot's 1946 doctoral thesis on how chess masters think.

In January 1956, walking into a graduate class at Carnegie Tech, he announced: “*Over Christmas, Al Newell and I invented a thinking machine.*” The piece of software they wrote is now seen as one of the foundations of artificial intelligence. And yet he was, by all accounts, a personable, warm man. His student Fernand Gobet wrote that Simon’s informal style made you sometimes forget you were talking to one of the greatest scholars of the twentieth century.

In 1962 Simon published *The Architecture of Complexity*, and, like this chapter, it opens with the parable of Tempus and Hora. Simon’s central claim was that all sufficiently complex systems have to function like Hora to survive. Life. Evolution. Cities. Companies. Any system complicated enough that you would consider calling it complex. He had a name for the property such systems share – the property of being built from semi-autonomous, independent sub-assemblies of the kind Hora uses on his bench. He called it *near-decomposability*.

Near-decomposable means this. Inside a module, interactions are strong and fast. Between modules, interactions are weak and slow. Parts within a sub-assembly bond tightly; sub-assemblies connect loosely. The asymmetry – strong-inside, weak-between – is what lets each level run its own business without being swamped by the others. Such a system is not *completely* decomposable. That would mean your cells have nothing to do with you. It is *near-decomposable*. Your cells communicate a lot more with the mitochondria inside of them than with you, but fortunately they listen enough to make you function as a human being.

The cell is just one example. A microchip is also near-decomposable: what happens inside the chip package is millions of times faster than what crosses the circuit board. A city neighbourhood is near-decomposable: you interact with your block several times a day, with the rest of the city occasionally, with the country a few times a year.

The Architecture of Complexity was the literary way to describe what Simon had shown one year earlier, with Albert Ando, mathematically. In this 1961 paper, Ando and Simon proved that in a near-decomposable system, fast local equilibria reach steady state inside their modules before the slow cross-module dynamics can interfere. The combinatorial nightmare in which every component is coupled to every other collapses into something a designer, or evolution, or a control engineer can actually reason about. Anything you can solve locally you should solve locally, partly because that is very fast. Anything you cannot solve locally you hand up a level – but this does tend to slow things down and create additional overhead.

So the parable of the two watchmakers is not merely intuition or story. It is something Simon had actually proved mathematically a year earlier.

The grid we currently run is the opposite. Every device, in principle, connects to every other. Every fault, every imbalance, every price signal propagates instantly through the whole thing. That is precisely the combinatorial nightmare that Simon and Ando warned about. They proved sixty years ago already that the current architecture of the grid is both more complicated and more vulnerable than it needs to be.

3. Koestler Who Coined the Term Holons

Although Simon described the mechanism, the word for it that we remember comes from [Arthur Koestler](#). Koestler was certainly [not a saint](#). When I think of him, I think of him first and foremost as a survivor of tyrannical monolithic systems – personally, at close range, under sentence of death more than once.

At the beginning of the Second World War, in Spain, he was arrested as a Comintern agent while reporting on the civil war, and condemned to death. The people in the cell next to him were shot. He only survived because one of Franco's ace pilots arranged a prisoner exchange for his wife. In 1939 he was interned as an undesirable alien in the concentration camp at [Le Vernet](#). In September 1940 he was so certain that he would die, horribly, at the hands of the Gestapo, that together with his friend [Walter Benjamin](#) he arranged for suicide tablets – which killed his friend, but somehow he survived. During that time he wrote [Darkness at Noon](#), about a victim broken by totalitarian Stalinist thought. And although you have probably never heard of it, it later led George Orwell to write *Animal Farm* and *Nineteen Eighty-Four*. So this is a man who knew what oppressive totalitarian systems looked like. And I can't help but read [The Ghost in the Machine](#) through this lens. He wrote it much later, in 1967, and it is not a political manifesto at all, unless you know this background. It is also a way to extend the mechanism that Herbert Simon had so brilliantly discovered, and to embed it in the social sciences. It is in this book that he coins the word *holon*, from the Greek *holos*, whole, with the suffix *-on* that physics already uses for particles like proton, neutron and photon.

A holon is simultaneously a whole in itself and a part of something larger. A cell is a holon inside a kidney, which is a holon inside a body, which is a holon inside an ecosystem. A home is a holon inside a street, inside a neighbourhood, inside a grid.

For me, Koestler was not merely renaming what Simon had described. He was seeing Simon through the eyes of a man who had tried to take his own life because of what monolithic systems tried to do to him. For him this was not just theory. It was lived experience. He did not only want systems that were more efficient and more resilient. He wanted systems that made us free.

4. Evolution Uses Holons

By now you might have noticed that the entire previous chapter was basically about giving you examples of how monolithic systems fail and how holons survive. The mitochondrion is a holon inside a cell. The cell is a holon inside the tissue. The tissue is a holon inside the organ. The organ lives inside a human being. Human beings form organisations, countries and in the end humanity. Holons make sure failure isn't catastrophic. An infected kidney does not infect the liver. A person dying doesn't end civilization. And life doesn't stop at making architecture robust. Life uses it to improve – as, for example, with the lodgepole pines that use fire to get the upper hand.

In 1996 the evolutionary biologists [Günter Wagner](#) and [Lee Altenberg](#) formulated further support for Simon's claim – that evolution needs holonic systems to increase the complexity of life.

If your organism is modular, a useful mutation in one trait – a slightly better-placed eye, a slightly more efficient enzyme, a wing feather that catches the air a little more cleanly – does not automatically scramble everything else. You can evolve the eye without breaking the kidney. If your organism is a monolith, with everything coupled to everything else, every small adjustment has a fair chance of being catastrophic. Evolution cannot get a foothold. The lineage does not persist.

Which, when you think about it, is a rather striking claim. Evolution itself – the single mechanism responsible for every living thing you have ever met, including yourself – does not work on monolithic systems. It *requires* holons. Three and a half billion years of the most ruthless optimisation process in the known universe, running continuously, on every species, in every niche, on every continent, in oceans, in deserts, in hot springs and ice cores and your own gut, and the verdict is unanimous: monoliths die, holons persist.

When we build a grid out of holons, we are not inventing a new architecture. We are adopting the one life on this planet has been stress-testing since before there was oxygen in the air.

5. The Internet Uses Holons

In 1984, three colleagues at MIT – [Jerome Saltzer](#), [David Reed](#) and [David Clark](#) – published a paper called *End-to-End Arguments in System Design*. Compressed into a sentence, its argument was: keep the network itself simple and stupid. Put the intelligence at the edges, close to the users. That is exactly what we are talking about when we are talking about home energy management systems, as you will meet in later chapters.

The network does not need to know what is flowing through it, let alone coordinate it all. This generic architecture made it possible to implement email, video, peer-to-peer, streaming and every application nobody had yet thought of. It could all be deployed without rebuilding the network underneath. If we apply this to the grid we can introduce local flexibility and energy communities everywhere, and we don't need any central organisation to keep them in check.

Of the three authors, the one who became the public face of the philosophy is [David Clark](#). From 1981 to 1989 he served as Chief Protocol Architect of the Internet – a title nobody has held since, and one that sounds, in retrospect, slightly fantastical. In [July 1992](#), in a plenary talk at the 24th Internet Engineering Task Force meeting, he compressed the entire design philosophy of the internet into fourteen words:

We reject kings, presidents, and voting. We believe in rough consensus and running code.

That sentence is one of the most quoted in the history of computing. It is also one of the most political things an engineer has ever said out loud, and it says a great deal about who the internet was built by and who it was built for. It was built by people who distrusted monolithic authority and had chosen a profession where that distrust could be encoded into architecture.

The reason to hold up the internet specifically, rather than some older system, is that the internet was designed for the world the grid now lives in. When the grid was designed, the

five “inalienable truths” I listed in the last chapter – *expensive planned supply, no storage, centralised planning, passive consumers, and the grid as a copper plate* – were actually true. But that was over a hundred years ago. When the internet was designed, not one of them still held. Supply of bits was cheap and anyone could generate them. Storage was cheap. Computation was becoming essentially free, so planning could be distributed. Users sat at PCs and were anything but passive. And packet-switching had replaced the copper-plate ideal with a network that routed around any bottleneck. That made intelligence at the edges feasible.

The internet is what the grid would look like if it were built today, under modern constraints instead of constraints more than a century out of date.

6. Good Governance Uses Holons

In law and political science, the name for holons is *subsidiarity*.

Subsidiarity means that a decision should be made at the scale that matches the scale of its effects, and larger scales should step in only where the problem genuinely spills over. That is not because small is virtuous. It is first and foremost because people like to be free to make decisions themselves. That in turn is partly because they know their own situation and their own desires best, and partly for a host of psychological reasons that we will come back to in Part Three. Apart from that, it reduces overhead, reduces the chance of cascading failures, speeds up decision-making, and thus leads to an overall more efficient and resilient system.

Holons – or subsidiarity – has been a part of Catholic social teaching for quite a while. The principle was articulated implicitly in Leo XIII’s *Rerum Novarum* in 1891 and received its canonical formulation forty years later in Pius XI’s *Quadragesimo Anno*, the key paragraphs drafted by the Jesuit *Oswald von Nell-Breuning*. The word reached EU treaty law at Maastricht in 1992 and sits today in *Article 5(3) of the Treaty on European Union*. The Union acts only where the objectives of the proposed action cannot be sufficiently achieved at national, regional, or local level. That is the test. It is a good test, but from casual observation of Brussels, not always the one that gets applied.

Everybody who knows Dutch history also knows the principle very well. In 1848, while thrones fell across Europe like decorations off an over-decorated Christmas tree, King William II, proclaimed he *converted from conservative to liberal in a single night*. He then summoned *Johan Rudolph Thorbecke* to write him a constitution. Thorbecke had written the most thorough commentary on the existing constitution anyone had ever produced, and this had made him rather unpopular. But now he was the perfect man for the job. He also wrote two further statutes – the *Provinciewet* of 1850 and the *Gemeentewet* of 1851. These explicitly distributed competences across three layers of government: national, provincial and municipal. Each layer had its own mandate, each was answerable to its own electorate, and none could absorb the functions of the others. In Dutch this is called *het Huis van Thorbecke* – the home of Thorbecke – and it has stood for over 175 years.

E. F. Schumacher’s *Small Is Beautiful* (1973), written by the former Chief Economic Advisor to the UK National Coal Board, was explicitly framed in terms of Catholic subsidiarity.

American federalism. Community-scale economics. Food sovereignty. Cooperative movements. Different accents, same architectural claim. Power works better when it is nested and autonomy is respected at every level: power works better when it uses holons.

To social scientists, political theorists, governance researchers and practitioners who have defended subsidiarity, federalism, polycentric commons or community-scale economics against the centralising default of modern policy I say: the holon framework is what you have been advocating for all along. This book is an argument that the energy system should finally catch up with your field.

When I look at all the citizen energy communities – village wind cooperatives, neighbourhood solar collectives, community batteries, *energiegemeenschappen* in Dutch law, *Bürgerenergie* in Germany, or simply people smart-charging their car or using solar with batteries – I see holons.

7. Holons to Escape the Tragedy of the Commons

One of the most inspiring characters in the holonic tradition – even though she prefers the word *polycentric governance* – is [Elinor Ostrom](#) and we will introduce her in more detail in the next chapter.

Her evidence was field data, not theory. Irrigation channels in the Spanish *huertas*. Alpine pastures in the Swiss village of [Törbel](#), continuously governed by a written agreement signed in 1224. Village forests in Japan. Inshore fisheries in Maine and Turkey. Groundwater basins under Los Angeles. Over decades and dozens of cases, polycentric governance of shared resources, matched to the scale of the resource and run by the users themselves, outperformed both state monopoly and private ownership.

Ostrom was a political scientist, not an economist; all three of her degrees, including her 1965 UCLA doctorate, were in political science. She built her career over 47 years at Indiana University Bloomington, where in 1973 she and her husband Vincent founded a research centre they insisted on calling a *Workshop*, run without a hierarchy. In 2009 she received the [Nobel Prize in Economics](#). She is the only woman to have been awarded one.

Her work has quietly demolished one of the oldest assumptions in her adopted field – that common resources have to be either privatised or regulated from above to survive. They do not. Real communities in real places had been governing real shared resources for, in the Swiss case, 800 years, without either an emperor or an invisible hand, and doing it, measurably, better than either would have.

A grid is a shared physical resource. The case for organising it polycentrically is, at this point, not a hypothesis. It is an overdetermined conclusion.

8. The Grid Uses Holons, Well, Almost

Look at any electricity grid. Every low-voltage transformer serves its own customers, who interact with the rest of the world only through the medium-voltage layer above them. Medium-voltage substations interact with the rest of the world through high-voltage

transmission. The physical topology of every grid on Earth is a perfect holarchy. The copper is already nested.

What is missing is buffers that can decouple different holons, grid-forming inverters that can steer power flows, computers that can automate the decision-making process, and the will to share power with end users and let them take their own decisions.

So let's define energy holons. An *energy holon* is a semi-autonomous microgrid that can maintain its own balance locally and presents a simple interface to the level above. That level above is then provided with all the flexibility the holon has to offer, without actually having anything to say about how the demands and flexibility are internally coordinated. An energy holon can stay stable for a limited amount of time on its own, but tries to contribute to the larger system whenever it can.

In the engineering literature, such a holon – such a microgrid – is usually small, like a house, a facility or a small cluster. The holon concept generalises it to any scale – from a single home, to a street, a neighbourhood, a city, a province, a country, or even a continent.

The architecture is not theoretical. The US National Renewable Energy Laboratory has been operating [microgrid testbeds](#) since 2001. At the other end of the income distribution, the one-house holon is already lived reality for tens of millions of households. The 2024 [Off-Grid Solar Market Trends Report](#), a joint publication of the World Bank, ESMAP and GOGA, projects that 398 million people will have first-time electricity access through off-grid solar by 2030 under baseline scenarios. That is the population of the United States plus another hundred million – every one of them a working one-house holon, by necessity, because the grid never arrived.

The idea that the grid should be built using holons is not new. [Yoichi Kaya](#) proposed a “*holonic path in the energy system*” in 1987. [Nishikawa and colleagues](#) worked out the first engineering version in 1992. [Wu, Kotak and Fleetwood](#) extended it in 2004 and 2005. [Negeri, Baken and Popov](#) consolidated it between 2012 and 2013. In the Netherlands, [Mart van Bracht](#), then at Topsector Energie, brought the word into national energy-policy discourse in October 2020.

This book merely gathers the different lineages under one roof and presents them in an integrated way, using prose that does not require an engineering degree to read.

So now we know that holons win across every domain humans have cared enough to test – biology, the internet, governance, commons management. And we have a grid whose physical topology is already well suited. The remaining question is how you organise the grid into such a system of holons – such a system of semi-autonomous units – without a commander to give the orders. How can cooperation arise without a central plan. And that is what the next chapter is all about.